

Dongyu Wei

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[Google Scholar](#)

Research Summary

I build secure and high-performance wireless systems for distributed AI at the edge, unifying privacy-preserving distributed learning with cross-layer optimization. I use multi-agent RL and semantic communications to make edge training resilient, spectrum/energy-efficient, and trustworthy.

Education

- **University of Miami, Coral Gables, USA** Aug 2024 – Jun 2028 (Expected)
Ph.D. in Electrical and Computer Engineering
Supervisor: Prof. Mingzhe Chen (IEEE Senior Member)
- **University of Washington, Seattle, USA** Sep 2022 – Jun 2024
M.S. in Electrical and Computer Engineering
- **Huazhong University of Science and Technology, Wuhan, China** Sep 2018 – Jun 2022
B.E. in Electrical Engineering

Research Experience

- **Research Assistant**, University of Miami, FL, USA Aug 2024 – Present
 - Conduct research on wireless communication systems and secure distributed learning frameworks.
 - Design and implement algorithms to enhance spectrum efficiency and enable privacy-preserving AI model training.
 - Evaluate system performance through large-scale simulations and data analysis using Python and MATLAB.
 - Collaborate with academic researchers to disseminate findings through peer-reviewed IEEE conferences and journals.

Publications

Peer-reviewed Journal and Conference Papers (Accepted/Published)

1. **D. Wei**, X. Xu, S. Mao, and M. Chen, “Optimizing Communication and Device Clustering for Clustered Federated Learning with Differential Privacy,” in *IEEE Transactions on Mobile Computing*, 2025.
2. **D. Wei**, L. Cao, L. Zhang, X. Gao, and H. Yin, “Non-Primary Channel Access in IEEE 802.11 UHR: Comprehensive Analysis and Evaluation,” in *IEEE 100th Vehicular Technology Conference (VTC-Fall)*, Washington, DC, USA, Oct. 2024.
3. **D. Wei**, L. Cao, L. Zhang, X. Gao, and H. Yin, “Optimized Non-Primary Channel Access Design in IEEE 802.11bn,” in *IEEE Global Communications Conference (GLOBECOM)*, Cape Town, South Africa, Dec. 2024.
4. **D. Wei**, H. Yu, Y. Liu, S. Mao, and M. Chen, “Joint Optimization of Communication and Device Clustering for Secure Clustered Federated Learning,” in *IEEE International Conference on Communications (ICC)*, Montreal, Canada, Jun. 2025.

5. **D. Wei**, Y. Liu, and M. Chen, “Joint Optimization of Model Splitting and Device Task Assignment for Private Multi-hop Split Learning,” in *IEEE Conference on Information Sciences and Systems (CISS)*, Baltimore, MD, USA, Mar. 2025.
6. X. Xu, H. Xu, **D. Wei**, W. Saad, M. Bennis, and M. Chen, “Fluid Antenna System and Other Next-Generation Reconfigurable Antenna Systems for Wireless Communications,” in *IEEE Journal on Selected Areas in Communications*, 2025.
7. L. Zhang, L. Cao, **D. Wei**, M. Chen, Z. Chen, and S. Cui, “Cross-Layer Channel Sounding Optimization Towards Next-Gen Wi-Fi: From Model Driven to Data Driven,” in *IEEE Transactions on Wireless Communications*, 2025.
8. L. Zhang, L. Cao, **D. Wei**, and X. Gao, “Scheduling in IEEE 802.11be Multi-Link Operation for Low Latency and High Reliability,” in *IEEE Transactions on Green Communications and Networking*, 2025.
9. X. Gao, H. Yin, Y. Sun, **D. Wei**, X. Xu, H. Chen, W. Wu, and S. Cui, “Multi-Level Feature Transmission in Dynamic Channels: A Semantic Knowledge Base and Deep Reinforcement Learning-Enabled Approach,” in *IEEE Internet of Things Journal*, 2024.
10. L. Zhang, L. Cao, **D. Wei**, M. Chen, Z. Chen, and V. Sathya, “Cross-Layer Channel Sounding Optimization Towards Next-Gen Wi-Fi,” in *IEEE Consumer Communications & Networking Conference (CCNC)*, Las Vegas, NV, USA, Jan. 2026.
11. X. Xu, **D. Wei**, Y. Hu, Z. Yang, H. Guo, and M. Chen, “Joint Trajectory and Antenna Port Selection Optimization for Fluid Antenna System-enabled Resilient UAV Networks,” in *IEEE Global Communications Conference (GLOBECOM)*, Taipei, Taiwan, Dec. 2025.
12. S. Yang, **D. Wei**, H. Yu, Z. Yang, Y. Liu, and M. Chen, “Contrastive Language-Image Pre-Training Model based Semantic Communication Performance Optimization,” in *IEEE Global Communications Conference (GLOBECOM)*, Taipei, Taiwan, Dec. 2025.
13. X. Gao, Y. Sun, **D. Wei**, X. Xu, H. Chen, H. Yin, and S. Cui, “Learning for Semantic Knowledge Base-Guided Online Feature Transmission in Dynamic Channels,” in *IEEE International Conference on Communications (ICC)*, Denver, CO, USA, 2024.

Under Review / Revision

1. **D. Wei**, X. Xu, Y. Liu, H.V. Poor, and M. Chen, “Optimizing Model Splitting and Device Task Assignment for Deceptive Signal Assisted Multi-hop Split Learning,” under major revision for *IEEE Journal on Selected Areas in Communications*.
2. **D. Wei**, H. Yu, Y. Liu, S. Mao, and M. Chen, “Joint Optimization of Federated Learning and Unlearning over Wireless Networks,” submitted to *IEEE INFOCOM*, Tokyo, Japan, May 2026.
3. **D. Wei**, X. Xu, Y. Liu, W. Zhang, and M. Chen, “Optimizing Device Assignment and Sub-model Splitting for Resilient Multi-hop Split Learning,” submitted to *IEEE International Conference on Communications (ICC)*, Glasgow, Scotland, UK, Jun. 2026.
4. J. Yao, **D. Wei**, and M. Chen, “Joint Optimization of Movable Antenna Positions, Beamforming, and UAV Deployment for Sensing Resilient UAV Communications,” submitted to *IEEE Journal on Selected Areas in Communications*.
5. L. Zhang, L. Cao, **D. Wei**, M. Chen, Y. Jia, and V. Sathya, “Dual Model and Data Driven Optimization Toward Multi-AP Coordinated Spatial Reuse,” submitted to *IEEE Transactions on Wireless Communications*.
6. L. Zhang, L. Cao, Z. Chen, **D. Wei**, M. Chen, V. Sathya, and S. Mao, “Wi-Fi 8 Coordinated Beamforming: A Cross-layer Approach towards Optimized Access Point Cluster Formation,” submitted to *IEEE Transactions on Wireless Communications*.
7. D. Zhou, **D. Wei**, Y. Hu, S. Li, and M. Chen, “Semantic Communication Performance Optimization with Channel and Content Preference Feedbacks,” submitted to *IEEE International Conference on Communications (ICC)*, Glasgow, Scotland, UK, Jun. 2026.
8. L. Zhang, Y. Jia, L. Cao, **D. Wei**, M. Chen, and V. Sathya, “A Model Driven Optimization Toward Next-Generation Multi-AP Coordinated Spatial Reuse,” submitted to *IEEE International Conference*

- on Communications (ICC), Glasgow, Scotland, UK, Jun. 2026.
9. P. Li, L. Cao, L. Zhang, **D. Wei**, Y. Hu, and Q. Xie, "EPFL-REMNet: Efficient Personalized Federated Digital Twin Towards 6G Heterogeneous Radio Environment," submitted to *IEEE International Conference on Communications (ICC)*, Glasgow, Scotland, UK, Jun. 2026.
 10. L. Cao, W. Wang, Q. Xie, **D. Wei**, and L. Zhang, "SALT-V: Lightweight Authentication for 5G V2X Broadcasting," submitted to *IEEE International Conference on Communications (ICC)*, Glasgow, Scotland, UK, Jun. 2026.
 11. Z. Liu, L. Cao, L. Zhang, **D. Wei**, Y. Hu, and W. Wang, "LLP-V2X: Low Latency-Power Vehicular Networking Towards 6G V2X," submitted to *IEEE International Conference on Communications (ICC)*, Glasgow, Scotland, UK, Jun. 2026.

Skills and Services

- **Programming:** Matlab, Python, LaTeX, C/C++, Java
- **Languages:** English, Chinese
- **Reviewer:**
 - IEEE Journal on Selected Areas in Communications
 - IEEE Transactions on Mobile Computing
 - IEEE Transactions on Communications
 - IEEE Transactions on Cognitive Communications and Networking
 - IEEE Transactions on Dependable and Secure Computing
 - IEEE Transactions on Green Communications and Networking
 - IEEE Transactions on Machine Learning in Communications and Networking
 - IEEE Communications Magazine
 - IEEE Wireless Communications Letters
 - IEEE International Conference on Communications (ICC)
 - IEEE Global Communications Conference (GLOBECOM)
 - IEEE INFOCOM Workshops
 - IEEE Consumer Communications & Networking Conference (CCNC)
 - ACM ASIACCS Workshops
 - Elsevier Computer Networks (COMNET)
- **Professional Service:**
 - Technical Program Committee (TPC) Member, *IEEE International Conference on Communications (ICC) 2026*, Signal Processing for Communications Symposium
 - Technical Program Committee (TPC) Member, *IEEE International Conference on Communications (ICC) 2026*, Wireless Communications Symposium

References

- **Dr. Mingzhe Chen (Ph.D. Advisor)**
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Department of Electrical and Computer Engineering, University of Miami
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- **Dr. Shiwen Mao**
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